Survey (Set 1)

When developers add new test cases into an existing test suite, do the developers look at the existing test cases and then reuse different attributes of existing test cases (such as variable naming, comments, test case naming, set of source code methods, overall test case organization/structure etc.) to derive the new testcases?

In order to answer the above question, we have designed this survey. The survey shows 20 pairs of unit test cases collected from open source projects. For each pair the participant is requested to indicate to what extent a developer *may/may not* have used a specific attribute to write the new test case.

Please answer all the following questions to complete the study. Thanks in advance.

* Indicates required question

Example 1: Following is a pair of test cases where the first one is an existing test case and the later one is the new test case. Please indicate to what extent a developer may have used the following attributes while writing the new test cases where -
(5=High extent of use)
(1=Low extent of use)

```java
/**
 * Test for IO-141
 */
public void testCopyDirectoryToGrandChild() throws Exception {
  File grandParentDir = new File(getTestDirectory(), "grandparent");
  File parentDir = new File(grandParentDir, "parent");
  File childDir = new File(parentDir, "child");
  createFilesForTestCopyDirectory(grandParentDir, parentDir, childDir);
  long expectedCount = (LIST_WALKER.list(grandParentDir).size() * 2);
  long expectedSize = (FileUtils.sizeOfDirectory(grandParentDir) * 2);
  FileUtils.copyDirectory(grandParentDir, childDir);
  assertEquals(expectedCount, LIST_WALKER.list(grandParentDir).size());
  assertEquals(expectedSize, FileUtils.sizeOfDirectory(grandParentDir));
}

public void testCopyDirectoryFiltered() throws Exception {
  File grandParentDir = new File(getTestDirectory(), "grandparent");
  File parentDir = new File(grandParentDir, "parent");
  File childDir = new File(parentDir, "child");
  createFilesForTestCopyDirectory(grandParentDir, parentDir, childDir);
  NameFileFilter filter = new NameFileFilter(new String[] {
    "parent",
    "child",
    "file3.txt"
  });
  File destDir = new File(getTestDirectory(), "copydest");
  FileUtils.copyDirectory(grandParentDir, destDir, filter);
  List files = LIST_WALKER.list(destDir);
  assertEquals(3, files.size());
  assertEquals("parent", ((File) files.get(0)).getName());
  assertEquals("child", ((File) files.get(1)).getName());
  assertEquals("file3.txt", ((File) files.get(2)).getName());
}
```

1. To what extent the developer may have used the "**Testcase Naming Style**" *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

2. To what extent the developer may have used the **"Variable Naming Style**" (if any)

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

3. To what extent the developer may have used the **"Variable or Object Construction**"

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

4. To what extent the developer may have used the "**Comments**" (if any)

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

5. To what extent the developer may have used the "**Set of Source Code Methods Invoked**" *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

6. To what extent the developer may have used the "**Assertions (i.e. the attributes/methods being invoked on the asserted object**" (if any)

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

7. To what extent the developer may have used the "**Testcase Structure/Organization Style**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

8. Overall did the developer used the existing test case to write the new one? *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

9. Please indicate if any other attributes were used (not listed in above)

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Untitled Section

Example 2: Following is a pair of test cases where the first one is an existing test case and the later one is the new test case. Please indicate to what extent a developer may have used the following attributes while writing the new test cases where -
(5=High extent of use)
(1=Low extent of use)

```
"public void testFiles() throws Exception {
    // XXX: This test presumes the current working dir is the base dir of the source checkout.
    IOFileFilter filter = FileFileFilter.FILE;
    assertFiltering(filter, new File(""src/""), false);
    assertFiltering(filter, new File(""src/java/""), false);
    assertFiltering(filter, new File(""pom.xml""), true);
    assertFiltering(filter, new File(""imaginary""), false);
    assertFiltering(filter, new File(""imaginary/""), false);
    assertFiltering(filter, new File(""LICENSE.txt""), true);
}
public void testFileFilterUtils_and() throws Exception {
    IOFileFilter trueFilter = TrueFileFilter.INSTANCE;
    IOFileFilter falseFilter = FalseFileFilter.INSTANCE;
    assertFiltering(FileFilterUtils.and(trueFilter, trueFilter, trueFilter), new File(""foo.test""), true);
    assertFiltering(FileFilterUtils.and(trueFilter, falseFilter, trueFilter), new File(""foo.test""), false);
    assertFiltering(FileFilterUtils.and(falseFilter, trueFilter), new File(""foo.test""), false);
    assertFiltering(FileFilterUtils.and(falseFilter, falseFilter), new File(""foo.test""), false);
}"
```

10. To what extent the developer may have used the "**Testcase Naming Style**" *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

11. To what extent the developer may have used the **"Variable Naming Style**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

12. To what extent the developer may have used the **"Variable or Object Construction"** (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

13. To what extent the developer may have used the "**Comments**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

14. To what extent the developer may have used the "**Set of Source Code Methods Invoked**" *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

15. To what extent the developer may have used the "**Assertions (i.e. the attributes/methods being invoked on the asserted object**" (if any) *

Mark only one oval.

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◯	◯	◯	◯	◯

16. To what extent the developer may have used the "**Testcase Structure/Organization Style**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

17. Overall did the developer used the existing test case to write the new one? *

Mark only one oval.

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◯	◯	◯	◯	◯

18. Please indicate if any other attributes were used (not listed in above)

Untitled Section

Example 3: Following is a pair of test cases where the first one is an existing test case and the later one is the new test case. Please indicate to what extent a developer may have used the following attributes while writing the new test cases where -
(5=High extent of use)
(1=Low extent of use)

```java
public void testChecksumOnNullFile() throws Exception {
  try {
    FileUtils.checksum((File) null, new CRC32());
    fail();
  } catch (NullPointerException ex) {
    // expected
  }
}
public void testDeleteQuietlyForNull() {
  try {
    FileUtils.deleteQuietly(null);
  } catch (Exception ex) {
    fail(ex.getMessage());
  }
}
```

19. To what extent the developer may have used the "**Testcase Naming Style**" *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

20. To what extent the developer may have used the **"Variable Naming Style**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

21. To what extent the developer may have used the **"Variable or Object Construction"** (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

22. To what extent the developer may have used the "**Comments**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

23. To what extent the developer may have used the "**Set of Source Code Methods Invoked**" *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

24. To what extent the developer may have used the "**Assertions (i.e. the attributes/methods being invoked on the asserted object**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

25. To what extent the developer may have used the "**Testcase Structure/Organization Style**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

26. Overall did the developer used the existing test case to write the new one? *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

27. Please indicate if any other attributes were used (not listed in above)

Untitled Section

Example 4: Following is a pair of test cases where the first one is an existing test case and the later one is the new test case. Please indicate to what extent a developer may have used the following attributes while writing the new test cases where -
(5=High extent of use)
(1=Low extent of use)

```
public void testWriteStringToFile1() throws Exception {
    File file = new File(getTestDirectory(), ""write.txt"");
    FileUtils.writeStringToFile(file, ""Hello /u1234"", ""UTF8"");
    byte[] text = ""Hello /u1234"".getBytes(""UTF8"");
    assertEqualContent(text, file);
}
public void testWriteCharSequence1() throws Exception {
    File file = new File(getTestDirectory(), ""write.txt"");
    FileUtils.write(file, ""Hello /u1234"", ""UTF8"");
    byte[] text = ""Hello /u1234"".getBytes(""UTF8"");
    assertEqualContent(text, file);
}
```

28. To what extent the developer may have used the "**Testcase Naming Style**" *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

29. To what extent the developer may have used the **"Variable Naming Style**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

30. To what extent the developer may have used the **"Variable or Object Construction**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

31. To what extent the developer may have used the "**Comments**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

32. To what extent the developer may have used the "**Set of Source Code Methods Invoked**" *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

33. To what extent the developer may have used the "**Assertions (i.e. the attributes/methods being invoked on the asserted object**" (if any) *

Mark only one oval.

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◯	◯	◯	◯	◯

34. To what extent the developer may have used the "**Testcase Structure/Organization Style**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

35. Overall did the developer used the existing test case to write the new one? *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

36. Please indicate if any other attributes were used (not listed in above)

Untitled Section

Example 5: Following is a pair of test cases where the first one is an existing test case and the later one is the new test case. Please indicate to what extent a developer may have used the following attributes while writing the new test cases where -
(5=High extent of use)
(1=Low extent of use)

```java
/**
 * Test the comparator array sort.
 */
public void testSortArray() {
    File[] files = new File[3];
    files[0] = equalFile1;
    files[1] = moreFile;
    files[2] = lessFile;
    comparator.sort(files);
    assertTrue("equal", files[0] == lessFile);
    assertTrue("less", files[1] == equalFile1);
    assertTrue("more", files[2] == moreFile);
}
/**
 * Test comparator array sort is null safe.
 */
public void testSortArrayNull() {
    assertNull(comparator.sort((File[]) null));
}
```

37. To what extent the developer may have used the "**Testcase Naming Style**" *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

38. To what extent the developer may have used the **"Variable Naming Style**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

39. To what extent the developer may have used the **"Variable or Object Construction"** (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

40. To what extent the developer may have used the "**Comments**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

41. To what extent the developer may have used the "**Set of Source Code Methods Invoked**" *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

42. To what extent the developer may have used the "**Assertions (i.e. the attributes/methods being invoked on the asserted object**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

43. To what extent the developer may have used the "**Testcase Structure/Organization Style**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

44. Overall did the developer used the existing test case to write the new one? *

Mark only one oval.

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◯	◯	◯	◯	◯

45. Please indicate if any other attributes were used (not listed in above)

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Untitled Section

Example 6: Following is a pair of test cases where the first one is an existing test case and the later one is the new test case. Please indicate to what extent a developer may have used the following attributes while writing the new test cases where -
(5=High extent of use)
(1=Low extent of use)

```java
public void testReadSwappedDouble() throws IOException {
    byte[] bytes = new byte[] { 0x08, 0x07, 0x06, 0x05, 0x04, 0x03, 0x02, 0x01 };
    double d1 = Double.longBitsToDouble(0x0102030405060708L);
    double d2 = EndianUtils.readSwappedDouble(bytes, 0);
    assertEquals(d1, d2, 0.0);
    ByteArrayInputStream input = new ByteArrayInputStream(bytes);
    assertEquals(d1, EndianUtils.readSwappedDouble(input), 0.0);
}
public void testEOFException() throws IOException {
    ByteArrayInputStream input = new ByteArrayInputStream(new byte[] {});
    try {
        EndianUtils.readSwappedDouble(input);
        fail("Expected EOFException");
    } catch (EOFException e) {
        // expected
    }
}
```

46. To what extent the developer may have used the "**Testcase Naming Style**" *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

47. To what extent the developer may have used the **"Variable Naming Style**" (if any) *

Mark only one oval.

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◯	◯	◯	◯	◯

48. To what extent the developer may have used the **"Variable or Object Construction"** (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

49. To what extent the developer may have used the "**Comments**" (if any) *

Mark only one oval.

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◯	◯	◯	◯	◯

50. To what extent the developer may have used the "**Set of Source Code Methods Invoked**" *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

51. To what extent the developer may have used the "**Assertions (i.e. the attributes/methods being invoked on the asserted object**" (if any) *

Mark only one oval.

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◯	◯	◯	◯	◯

52. To what extent the developer may have used the "**Testcase Structure/Organization Style**" (if any) *

Mark only one oval.

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◯	◯	◯	◯	◯

53. Overall did the developer used the existing test case to write the new one? *

Mark only one oval.

1	2	3	4	5
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54. Please indicate if any other attributes were used (not listed in above)

Untitled Section

Example 7: Following is a pair of test cases where the first one is an existing test case and the later one is the new test case. Please indicate to what extent a developer may have used the following attributes while writing the new test cases where -
(5=High extent of use)
(1=Low extent of use)

```java
/**
 * Test Filtering and limit to depth 5
 */
public void testFilterAndLimitD() {
    List<File> results = new TestFileFinder(dirsAndFilesFilter, 5).find(javaDir);
    assertEquals(""[D] Result Size"", (1 + dirs.length + ioFiles.length), results.size());
    assertTrue(""[D] Start Dir"", results.contains(javaDir));
    checkContainsFiles(""[D] Dir"", dirs, results);
    checkContainsFiles(""[D] File"", ioFiles, results);
}
/**
 * Test Filtering
 */
public void testFilterString() {
    List<String> results = new TestFileFinderString(dirsAndFilesFilter, -1).find(javaDir);
    assertEquals(""Result Size"", (outputFiles.length + ioFiles.length), results.size());
    checkContainsString(""IO File"", ioFiles, results);
    checkContainsString(""Output File"", outputFiles, results);
}
```

55. To what extent the developer may have used the "**Testcase Naming Style**" *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

56. To what extent the developer may have used the **"Variable Naming Style**" (if any) *

Mark only one oval.

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◯	◯	◯	◯	◯

57. To what extent the developer may have used the **"Variable or Object Construction"** (if any) *

Mark only one oval.

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◯	◯	◯	◯	◯

58. To what extent the developer may have used the "**Comments**" (if any) *

Mark only one oval.

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◯	◯	◯	◯	◯

59. To what extent the developer may have used the "**Set of Source Code Methods Invoked**" *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

60. To what extent the developer may have used the "**Assertions (i.e. the attributes/methods being invoked on the asserted object**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

61. To what extent the developer may have used the "**Testcase Structure/Organization Style**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

62. Overall did the developer used the existing test case to write the new one? *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

63. Please indicate if any other attributes were used (not listed in above)

Untitled Section

Example 8: Following is a pair of test cases where the first one is an existing test case and the later one is the new test case. Please indicate to what extent a developer may have used the following attributes while writing the new test cases where -
(5=High extent of use)
(1=Low extent of use)

```java
/**
 * Test writing Null String
 */
public void testNullString() {
    ProxyWriter proxy = new ProxyWriter(new NullWriter());
    try {
        proxy.write((String) null);
    } catch (Exception e) {
        fail(""Writing null String threw "" + e);
    }
    try {
        proxy.write((String) null, 0, 0);
    } catch (Exception e) {
        fail(""Writing null String threw "" + e);
    }
}
/**
 * Test Writing a String
 */
public void testWriteString() {
    StringBuilderWriter writer = new StringBuilderWriter();
    ProxyWriter proxy = new ProxyWriter(writer);
    try {
        proxy.write(""ABC"");
    } catch (Exception e) {
        fail(""Writing String threw "" + e);
    }
    assertEquals(""ABC"", writer.toString());
}
```

64. To what extent the developer may have used the "**Testcase Naming Style**" *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

65. To what extent the developer may have used the **"Variable Naming Style"** (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

66. To what extent the developer may have used the **"Variable or Object Construction"** (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

67. To what extent the developer may have used the "**Comments**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

68. To what extent the developer may have used the "**Set of Source Code Methods Invoked**" *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

69. To what extent the developer may have used the "**Assertions (i.e. the attributes/methods being invoked on the asserted object**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

70. To what extent the developer may have used the "**Testcase Structure/Organization Style**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

71. Overall did the developer used the existing test case to write the new one? *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

72. Please indicate if any other attributes were used (not listed in above)

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Untitled Section

Example 9: Following is a pair of test cases where the first one is an existing test case and the later one is the new test case. Please indicate to what extent a developer may have used the following attributes while writing the new test cases where -
(5=High extent of use)
(1=Low extent of use)

```
"@Test
public void testSetLength() {
    final int length = 99;
    RandomStringGenerator generator = new RandomStringGenerator.Builder().build();
    String str = generator.generate(length);
    assertEquals(length, codePointLength(str));
}
@Test
public void testGenerateMinMaxLength() {
    final int minLength = 0;
    final int maxLength = 3;
    RandomStringGenerator generator = new RandomStringGenerator.Builder().build();
    String str = generator.generate(minLength, maxLength);
    assertThat(codePointLength(str), allOf(greaterThanOrEqualTo(0), lessThanOrEqualTo(3)));
}
```

73. To what extent the developer may have used the "**Testcase Naming Style**" *

Mark only one oval.

1	2	3	4	5
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74. To what extent the developer may have used the **"Variable Naming Style**" (if any) *

Mark only one oval.

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75. To what extent the developer may have used the **"Variable or Object Construction"** (if any) *

Mark only one oval.

1	2	3	4	5
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76. To what extent the developer may have used the "**Comments**" (if any) *

Mark only one oval.

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77. To what extent the developer may have used the "**Set of Source Code Methods Invoked**" *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

78. To what extent the developer may have used the "**Assertions (i.e. the attributes/methods being invoked on the asserted object**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

79. To what extent the developer may have used the "**Testcase Structure/Organization Style**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

80. Overall did the developer used the existing test case to write the new one? *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

81. Please indicate if any other attributes were used (not listed in above)

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Untitled Section

Example 10: Following is a pair of test cases where the first one is an existing test case and the later one is the new test case. Please indicate to what extent a developer may have used the following attributes while writing the new test cases where -
(5=High extent of use)
(1=Low extent of use)

```java
public void test_checkEndsWith_case() throws Exception {
    assertEquals(true, IOCase.SENSITIVE.checkEndsWith(""ABC"", ""BC""));
    assertEquals(false, IOCase.SENSITIVE.checkEndsWith(""ABC"", ""Bc""));
    assertEquals(true, IOCase.INSENSITIVE.checkEndsWith(""ABC"", ""BC""));
    assertEquals(true, IOCase.INSENSITIVE.checkEndsWith(""ABC"", ""Bc""));
    assertEquals(true, IOCase.SYSTEM.checkEndsWith(""ABC"", ""BC""));
    assertEquals(WINDOWS, IOCase.SYSTEM.checkEndsWith(""ABC"", ""Bc""));
}
public void test_checkIndexOf_case() throws Exception {
    assertEquals(1, IOCase.SENSITIVE.checkIndexOf(""ABC"", 0, ""BC""));
    assertEquals(-1, IOCase.SENSITIVE.checkIndexOf(""ABC"", 0, ""Bc""));
    assertEquals(1, IOCase.INSENSITIVE.checkIndexOf(""ABC"", 0, ""BC""));
    assertEquals(1, IOCase.INSENSITIVE.checkIndexOf(""ABC"", 0, ""Bc""));
    assertEquals(1, IOCase.SYSTEM.checkIndexOf(""ABC"", 0, ""BC""));
    assertEquals(WINDOWS ? 1 : -1, IOCase.SYSTEM.checkIndexOf(""ABC"", 0, ""Bc""));
}
```

82. To what extent the developer may have used the "**Testcase Naming Style**" *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

83. To what extent the developer may have used the **"Variable Naming Style**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

84. To what extent the developer may have used the **"Variable or Object Construction"** (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

85. To what extent the developer may have used the "**Comments**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

86. To what extent the developer may have used the "**Set of Source Code Methods Invoked**" *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

87. To what extent the developer may have used the "**Assertions (i.e. the attributes/methods being invoked on the asserted object**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

88. To what extent the developer may have used the "**Testcase Structure/Organization Style**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

89. Overall did the developer used the existing test case to write the new one? *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

90. Please indicate if any other attributes were used (not listed in above)

_____

Untitled Section

Example 11: Following is a pair of test cases where the first one is an existing test case and the later one is the new test case. Please indicate to what extent a developer may have used the following attributes while writing the new test cases where -
(5=High extent of use)
(1=Low extent of use)

```
"/* Note: This test case tests the simplest functionality of
     * ObjectInputStream.  IF we really wanted to test ClassLoaderObjectInputStream
     * we would probably need to create a transient Class Loader. -TO
     */
public void testExpected() throws Exception {
    ByteArrayOutputStream baos = new ByteArrayOutputStream();
    ObjectOutputStream oos = new ObjectOutputStream(baos);
    oos.writeObject(Boolean.FALSE);
    InputStream bais = new ByteArrayInputStream(baos.toByteArray());
    ClassLoaderObjectInputStream clois =
        new ClassLoaderObjectInputStream(getClass().getClassLoader(), bais);
    Boolean result = (Boolean) clois.readObject();
    assertTrue(!result.booleanValue());
}
public void testResolveProxyClass() throws Exception {
    ByteArrayOutputStream baos = new ByteArrayOutputStream();
    ObjectOutputStream oos = new ObjectOutputStream(baos);
    oos.writeObject(Boolean.FALSE);
    InputStream bais = new ByteArrayInputStream(baos.toByteArray());
    ClassLoaderObjectInputStream clois =
        new ClassLoaderObjectInputStream(getClass().getClassLoader(), bais);
    String[] interfaces = new String[] { Comparable.class.getName() };
    Class<?> result = clois.resolveProxyClass(interfaces);
    assertTrue(""Assignable"", Comparable.class.isAssignableFrom(result));
}"
```

91. To what extent the developer may have used the "**Testcase Naming Style**" *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

92. To what extent the developer may have used the **"Variable Naming Style**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

93. To what extent the developer may have used the **"Variable or Object Construction"** (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

94. To what extent the developer may have used the "**Comments**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

95. To what extent the developer may have used the "**Set of Source Code Methods Invoked**" *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

96. To what extent the developer may have used the "**Assertions (i.e. the attributes/methods being invoked on the asserted object**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

97. To what extent the developer may have used the "**Testcase Structure/Organization Style**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

98. Overall did the developer used the existing test case to write the new one? *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

99. Please indicate if any other attributes were used (not listed in above)

_____

Untitled Section

Example 12: Following is a pair of test cases where the first one is an existing test case and the later one is the new test case. Please indicate to what extent a developer may have used the following attributes while writing the new test cases where -
(5=High extent of use)
(1=Low extent of use)

```java
/**
 * Tests constructor.
 */
@Test
public void testConstructorMapPrefixSuffix() {
    final Map<String, String> map = new HashMap<>();
    map.put(""name"", ""commons"");
    final StrSubstitutor sub = new StrSubstitutor(map, ""<"", ""&gt;"")
    assertEquals(""Hi < commons"", sub.replace(""Hi $< <name>""));
}
@Test
public void testReplaceInTakingStringBuilderWithNull() {
    Map<String, Object> map = new HashMap<>();
    StrSubstitutor strSubstitutor =
            new StrSubstitutor(map, """", """", 'T', ""K+<'f"");
    assertFalse(strSubstitutor.replaceIn((StringBuilder) null));
}
```

100. To what extent the developer may have used the "**Testcase Naming Style**" *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

101. To what extent the developer may have used the **"Variable Naming Style**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

102. To what extent the developer may have used the **"Variable or Object Construction**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

103. To what extent the developer may have used the "**Comments**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

104. To what extent the developer may have used the "**Set of Source Code Methods Invoked**" *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

105. To what extent the developer may have used the "**Assertions (i.e. the attributes/methods being invoked on the asserted object**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

106. To what extent the developer may have used the "**Testcase Structure/Organization Style**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

107. Overall did the developer used the existing test case to write the new one? *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

108. Please indicate if any other attributes were used (not listed in above)

_____

Untitled Section

Example 13: Following is a pair of test cases where the first one is an existing test case and the later one is the new test case. Please indicate to what extent a developer may have used the following attributes while writing the new test cases where -
(5=High extent of use)
(1=Low extent of use)

```java
public void testGetBOMFirstThenRead() throws Exception {
    byte[] data = new byte[] { 'A', 'B', 'C' };
    BOMInputStream in = new BOMInputStream(createDataStream(data, true));
    assertEquals("getBOM", ByteOrderMark.UTF_8, in.getBOM());
    assertTrue("hasBOM()", in.hasBOM());
    assertTrue("hasBOM(UTF-8)", in.hasBOM(ByteOrderMark.UTF_8));
    assertEquals('A', in.read());
    assertEquals('B', in.read());
    assertEquals('C', in.read());
    assertEquals(-1, in.read());
}
public void testGetBOMFirstThenReadInclude() throws Exception {
    byte[] data = new byte[] { 'A', 'B', 'C' };
    BOMInputStream in = new BOMInputStream(createDataStream(data, true), true);
    assertTrue("hasBOM()", in.hasBOM());
    assertTrue("hasBOM(UTF-8)", in.hasBOM(ByteOrderMark.UTF_8));
    assertEquals("getBOM", ByteOrderMark.UTF_8, in.getBOM());
    assertEquals(0xEF, in.read());
    assertEquals(0xBB, in.read());
    assertEquals(0xBF, in.read());
    assertEquals('A', in.read());
    assertEquals('B', in.read());
    assertEquals('C', in.read());
    assertEquals(-1, in.read());
}
```

109. To what extent the developer may have used the "**Testcase Naming Style**" *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

110. To what extent the developer may have used the **"Variable Naming Style"** (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

111. To what extent the developer may have used the **"Variable or Object Construction**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

112. To what extent the developer may have used the "**Comments**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

113. To what extent the developer may have used the "**Set of Source Code Methods Invoked**" *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

114. To what extent the developer may have used the "**Assertions (i.e. the attributes/methods being invoked on the asserted object**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

115. To what extent the developer may have used the "**Testcase Structure/Organization Style**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

116. Overall did the developer used the existing test case to write the new one? *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

117. Please indicate if any other attributes were used (not listed in above)

_____

Untitled Section

Example 14: Following is a pair of test cases where the first one is an existing test case and the later one is the new test case. Please indicate to what extent a developer may have used the following attributes while writing the new test cases where -
(5=High extent of use)
(1=Low extent of use)

```
"/**
 * Test {@link ByteOrderMark#getCharsetName()}
 */
public void testCharsetName() {
    assertEquals(""test1 name"", ""test1"", TEST_BOM_1.getCharsetName());
    assertEquals(""test2 name"", ""test2"", TEST_BOM_2.getCharsetName());
    assertEquals(""test3 name"", ""test3"", TEST_BOM_3.getCharsetName());
}
/**
 * Tests that {@link ByteOrderMark#getCharsetName()}
 * can be loaded as a {@link java.nio.charset.Charset} as advertised.
 */
public void testConstantCharsetNames() {
    assertNotNull(Charset.forName(ByteOrderMark.UTF_16BE.getCharsetName()));
    assertNotNull(Charset.forName(ByteOrderMark.UTF_16LE.getCharsetName()));
    assertNotNull(Charset.forName(ByteOrderMark.UTF_8.getCharsetName()));
}"
```

118. To what extent the developer may have used the "**Testcase Naming Style**" *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

119. To what extent the developer may have used the **"Variable Naming Style**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

120. To what extent the developer may have used the **"Variable or Object Construction**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

121. To what extent the developer may have used the "**Comments**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

122. To what extent the developer may have used the "**Set of Source Code Methods Invoked**" *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

123. To what extent the developer may have used the "**Assertions (i.e. the attributes/methods being invoked on the asserted object**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

124. To what extent the developer may have used the "**Testcase Structure/Organization Style**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

125. Overall did the developer used the existing test case to write the new one? *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

126. Please indicate if any other attributes were used (not listed in above)

_____

Untitled Section

Example 15: Following is a pair of test cases where the first one is an existing test case and the later one is the new test case. Please indicate to what extent a developer may have used the following attributes while writing the new test cases where -
(5=High extent of use)
(1=Low extent of use)

```java
@Test
public void testByAttribute() {
    String h = "<div title=foo /><div title=bar /><div style=qux />";
    Document doc = Jsoup.parse(h);
    Elements withTitle = doc.select("[title]");
    Elements foo = doc.select("[title=foo]");
    assertEquals(2, withTitle.size());
    assertEquals(1, foo.size());
}
@Test
public void caseInsensitive() {
    String h = "<div title=bar><div>";
    Document doc = Jsoup.parse(h);
    assertEquals(2, doc.select("DIV").size());
    assertEquals(1, doc.select("DIV[TITLE]").size());
    assertEquals(1, doc.select("DIV[TITLE=BAR]").size());
    assertEquals(0, doc.select("DIV[TITLE=BARBARELLA").size());
}
```

127. To what extent the developer may have used the "**Testcase Naming Style**" *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

128. To what extent the developer may have used the **"Variable Naming Style**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

129. To what extent the developer may have used the **"Variable or Object Construction**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

130. To what extent the developer may have used the "**Comments**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

131. To what extent the developer may have used the "**Set of Source Code Methods Invoked**" *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

132. To what extent the developer may have used the "**Assertions (i.e. the attributes/methods being invoked on the asserted object**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

133. To what extent the developer may have used the "**Testcase Structure/Organization Style**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

134. Overall did the developer used the existing test case to write the new one? *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

135. Please indicate if any other attributes were used (not listed in above)

Untitled Section

Example 16: Following is a pair of test cases where the first one is an existing test case and the later one is the new test case. Please indicate to what extent a developer may have used the following attributes while writing the new test cases where -
(5=High extent of use)
(1=Low extent of use)

```
@Test
public void testSha512HexInputStream() throws IOException {
    assertEquals(DigestUtils.sha512Hex(testData), DigestUtils.sha512Hex(new
        ByteArrayInputStream(testData)));
}
@Test
public void testSha256HexInputStream() throws IOException {
    assertEquals(DigestUtils.sha256Hex(testData), DigestUtils.sha256Hex(new
        ByteArrayInputStream(testData)));
}"
```

136. To what extent the developer may have used the "**Testcase Naming Style**" *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

137. To what extent the developer may have used the **"Variable Naming Style**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

138. To what extent the developer may have used the **"Variable or Object Construction**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

139. To what extent the developer may have used the "**Comments**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

140. To what extent the developer may have used the "**Set of Source Code Methods Invoked**" *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

141. To what extent the developer may have used the "**Assertions (i.e. the attributes/methods being invoked on the asserted object**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

142. To what extent the developer may have used the "**Testcase Structure/Organization Style**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

143. Overall did the developer used the existing test case to write the new one? *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

144. Please indicate if any other attributes were used (not listed in above)

Untitled Section

Example 17: Following is a pair of test cases where the first one is an existing test case and the later one is the new test case. Please indicate to what extent a developer may have used the following attributes while writing the new test cases where -
(5=High extent of use)
(1=Low extent of use)

```
"@Test
public void testEmptyFile() throws Exception {
    try (final CSVParser parser = CSVParser.parse("",
                                                  CSVFormat.DEFAULT)) {
        assertNull(parser.nextRecord());
    }
}
@Test
public void testEmptyFileHeaderParsing() throws Exception {
    try (final CSVParser parser = CSVParser.parse("",
                CSVFormat.DEFAULT.withFirstRecordAsHeader())) {
        assertNull(parser.nextRecord());
        assertTrue(parser.getHeaderNames().isEmpty());
    }
}
```

145. To what extent the developer may have used the "**Testcase Naming Style**" *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

146. To what extent the developer may have used the **"Variable Naming Style**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

147. To what extent the developer may have used the **"Variable or Object Construction**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

148. To what extent the developer may have used the "**Comments**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

149. To what extent the developer may have used the "**Set of Source Code Methods Invoked**" *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

150. To what extent the developer may have used the "**Assertions (i.e. the attributes/methods being invoked on the asserted object**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

151. To what extent the developer may have used the "**Testcase Structure/Organization Style**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

152. Overall did the developer used the existing test case to write the new one? *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

153. Please indicate if any other attributes were used (not listed in above)

Untitled Section

Example 18: Following is a pair of test cases where the first one is an existing test case and the later one is the new test case. Please indicate to what extent a developer may have used the following attributes while writing the new test cases where -
(5=High extent of use)
(1=Low extent of use)

```
@Test
public void descendant() {
    String h = "<div class=head><p class=first>Hello</p>
            <p>There</p></div><p>None</p>";
    Document doc = Jsoup.parse(h);
    Elements els = doc.select(".head p");
    assertEquals(2, els.size());
    assertEquals("Hello", els.get(0).text());
    assertEquals("There", els.get(1).text());
    Elements p = doc.select("p.first");
    assertEquals(1, p.size());
    assertEquals("Hello", p.get(0).text());
}
@Test
public void deeperDescendant() {
    String h = "<div class=head><p class=first>Hello</div>
        <div class=head><h1 class=first>Another</h1><p>Again</div>";
    Elements els = Jsoup.parse(h).select("div p .first");
    assertEquals(1, els.size());
    assertEquals("Hello", els.get(0).text());
}
```

154. To what extent the developer may have used the "**Testcase Naming Style**" *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

155. To what extent the developer may have used the **"Variable Naming Style**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

156. To what extent the developer may have used the **"Variable or Object Construction**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

157. To what extent the developer may have used the "**Comments**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

158. To what extent the developer may have used the "**Set of Source Code Methods Invoked**" *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

159. To what extent the developer may have used the "**Assertions (i.e. the attributes/methods being invoked on the asserted object**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

160. To what extent the developer may have used the "**Testcase Structure/Organization Style**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

161. Overall did the developer used the existing test case to write the new one? *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

162. Please indicate if any other attributes were used (not listed in above)

Untitled Section

Example 19: Following is a pair of test cases where the first one is an existing test case and the later one is the new test case. Please indicate to what extent a developer may have used the following attributes while writing the new test cases where -
(5=High extent of use)
(1=Low extent of use)

```
@Test
public void testBasic5() {
    final String input = ""a:b':c"";
    final StrTokenizer tok = new StrTokenizer(input, ':', '\'');
    assertEquals(""a"", tok.next());
    assertEquals(""b'"", tok.next());
    assertEquals(""c"", tok.next());
    assertFalse(tok.hasNext());
}
@Test
public void testDelimMatcher() {
    final String input = ""a/b\\c"";
    final StrMatcher delimMatcher =
            new StrMatcher.CharSetMatcher(new char[] { '/', '\\' });
    final StrTokenizer tok = new StrTokenizer(input, delimMatcher);
    assertEquals(""a"", tok.next());
    assertEquals(""b"", tok.next());
    assertEquals(""c"", tok.next());
    assertFalse(tok.hasNext());
}
```

163. To what extent the developer may have used the "**Testcase Naming Style**" *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

164. To what extent the developer may have used the **"Variable Naming Style**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

165. To what extent the developer may have used the **"Variable or Object Construction**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

166. To what extent the developer may have used the "**Comments**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

167. To what extent the developer may have used the "**Set of Source Code Methods Invoked**" *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

168. To what extent the developer may have used the "**Assertions (i.e. the attributes/methods being invoked on the asserted object**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

169. To what extent the developer may have used the "**Testcase Structure/Organization Style**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

170. Overall did the developer used the existing test case to write the new one? *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

171. Please indicate if any other attributes were used (not listed in above)

Untitled Section

Example 20: Following is a pair of test cases where the first one is an existing test case and the later one is the new test case. Please indicate to what extent a developer may have used the following attributes while writing the new test cases where -
(5=High extent of use)
(1=Low extent of use)

```
"@Test
public void testBasicEmpty1() {
    final String input = ""a  b c"";
    final StrTokenizer tok = new StrTokenizer(input);
    tok.setIgnoreEmptyTokens(false);
    assertEquals(""a"", tok.next());
    assertEquals("", tok.next());
    assertEquals(""b"", tok.next());
    assertEquals(""c"", tok.next());
    assertFalse(tok.hasNext());
}
@Test
public void testDelimString() {
    final String input = ""a##b##c"";
    final StrTokenizer tok = new StrTokenizer(input, ""##"");
    assertEquals(""a"", tok.next());
    assertEquals(""b"", tok.next());
    assertEquals(""c"", tok.next());
    assertFalse(tok.hasNext());
}"
```

172. To what extent the developer may have used the "**Testcase Naming Style**" *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

173. To what extent the developer may have used the **"Variable Naming Style**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

174. To what extent the developer may have used the **"Variable or Object Construction**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

175. To what extent the developer may have used the "**Comments**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

176. To what extent the developer may have used the "**Set of Source Code Methods Invoked**" *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

177. To what extent the developer may have used the "**Assertions (i.e. the attributes/methods being invoked on the asserted object**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

178. To what extent the developer may have used the "**Testcase Structure/Organization Style**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

179. Overall did the developer used the existing test case to write the new one? *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

180. Please indicate if any other attributes were used (not listed in above)

Demographic Information

181. What is you current post * Dropdown

Mark only one oval.

- PhD Student
- Masters Student
- Undergraduate Student
- Software Developer
- Others

182. How many years of programming experience you have? Dropdown

Mark only one oval.

- 0-1 year
- 1-3 years
- 3-5 years
- more than 5 years

183. Do you have experience in writing unit test cases?

Mark only one oval.

- Yes
- No
- Maybe

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